SMS Fraud Detection Using Machine Learning

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Abstract The term SMS stands for short message service. It is a message networking method that uses smartphones and cell phones. It is a text messaging system that lets smart phones to communicate with one another. SMS has risen in prominence in current history, with firms adapting and optimizing SMS to be used as a service tool mostly because of its numerous benefits. We are using a unique technique that uses data mining and machine learning method to identify and filter spam communications in this study. We analyzed the characteristics of spam SMS in depth and discovered ten criteria that may effectively distinguish short message (or messaging) service spam from ham. If some adjustments are needed to identify a new scam, they were made manually, either by applying changes to current algorithms or by inventing new algorithms. In this method, as the quantity of clients and data grows, so does the amount of human work, which may also effectively separate spam text messages and ham SMS messages. The random forest classification method, we proposed, is the methodology that produced 99.9% accuracy.

Keywords SMS spam · Machine learning · Data preprocessing · Networking

1 Introduction

A message is transmitted digitally via the short messaging service (SMS), which is one of the most widely used communication services. Telecommunication firms have cut the price of SMS services, which has resulted in increasing SMS usage. These increases recruited hackers, resulting in an SMS spam problem. Any unsolicited message sent from a user's device is referred to as spam. Advertisements, free

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services, rewards, and rewards are examples of spam communications. People prefer SMS messages to emails for communication because delivering SMS messages requires no Internet connection and is convenient and easy. The increased usage of text messaging and the issue of spams are becoming more prevalent. There seems to be a lot of security solutions available to mitigate the issue of SMS spam, although they are not yet developed. Many android applications exist on the play store to stop spam texts; however, due to lack of awareness, most people are unaware of them. Apart from applications, the existing monitoring solutions mostly concentrate on spam messages, as email is among the biggest problems, but even with the rise in the use of android platforms, SMS spam has become a big concern. Because mobile phones hold sensitive personal information such as card details, usernames, and passwords, SMS has been one of the cheapest ways to communicate and may be regarded as the easiest way to conduct phishing scams. Hackers are devising new ways of obtaining this data from smart phones, with SMS being one of the most straightforward. Type of phishing attack, or SMS-based hacking, is becoming increasingly prevalent these days. Phishing through SMS is becoming more common around the world, in which a person provides a malicious Web site by SMS and urges the recipient to click it, stealing important information from the recipient's smart phone. For identifying mobile fraud, there seems to be a variety of screening methods available, including smart cards, machine learning, fingerprints, matrix code scanner based, academic objectives, and identification based. SMS scammers may buy any contact information with just about any telephone number and use it to deliver spam messages, making it more difficult to track down the perpetrator. The top twenty spam detection area codes utilized by fraudsters were supplied by the US tango education center, in addition to the top ten SMS spam messages.

Cloudmark published a study in 2019 that detailed how hackers exploited SMS to deliver 876,000 spam messages as shown in Fig. 1. In 2020, the National Fraud Intelligence Bureau (NFIB) released a news report on most recent frauds, which has been examined by actions fraud. Phishers are attacking financial institution clients these days by delivering spam messages requesting for personal bank information, ATM pin number, and password. The major goal of our suggested strategy is to use machine

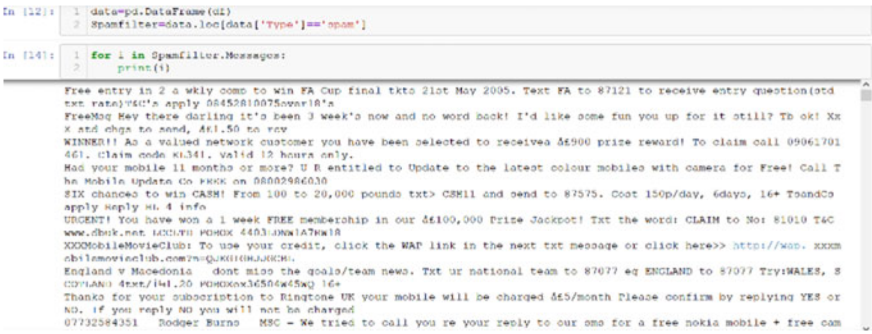


Fig. 1 Top five short message (or messaging) service spam messages

learning techniques to identify spam and ham SMS. For classification, we utilized a feature set of ten characteristics. These characteristics help distinguishing a spam SMS from a ham SMS. Machine learning techniques proved successful in avoiding zero-day threats and providing a high level of security in spam email filtering. The same method is being utilized for smart phones to eliminate short messaging spam; however, the characteristics of spam detection will differ from those of email spam since text messages are smaller and users use proper language for SMS.

2 Literature Review

There are a variety of text recognition text detection strategies accessible these days, such as blocking spam messages using mobile application and filter spam messages with classification techniques. In this paper, we will look at how to identify short messaging fraud applying machine learning approaches to filter spam messages depending on selecting features. El-Alfy and AlHasan [1] have suggested a short messaging filtering technique for both SMS and email. They looked at several approaches in order to come up with extracted features that would decrease difficulty. They utilized 13 characteristics, including URLs, probable spam terms, emotional icons, capital letters, wrinkly words, text data, JavaScript code, word structure, sender address, topic field, and junk mail domain, as well as two classification techniques, naive Bayes and support vector machine. Five email and SMS datasets were used to test their suggested model. For screening fraud communications, a messages subject structure (MTM) has been suggested by Jialin et al. [2]. To describe spam communications, the messaging predictive model (MTM) uses symbols, words, context terms, and subject terms and is predicated on a likelihood guess from latent semantic. They removed the scant difficulty by classifying SMS spam messages in to the unpredictable irregularity classes and afterward collecting all SMS phishing into a single document to detect keyword founder patterns using the k-means algorithm. The feature reweighting approach and the great word assault are two strategies for SMS spam detection that has been described by Chan et al. [3]. Both techniques concentrate on the text's size as well as the text's weight. The great word approach focuses on misleading the classifier's result by utilizing the fewest number of characters possible, whereas the characteristic reweighting technique has a novel parameterized function for resampling the dimensions. The experiment was assessed using two datasets: SMS and comments. Delany et al. [4] discussed the many techniques available for SMS spam filtering as well as the issues related with dataset gathering. They utilized ten groups to evaluate a huge dataset of SMS spam, including music, contests, dating, rewards, services, money, claim chatting, voice, and others. SMS spam messages were identified through information characteristics by Xu et al. [5]. For research experiment, they utilized two classification methods, SVM and k-nearest neighbors (KNN), as well as a feature set of factors, static, and temporal. They discovered that by integrating temporal and network characteristics SMS spam messages may be identified more precisely and efficiently. Furthermore, they discovered methods to filter SMS spam messages by utilizing characteristics that comprise chart and time series, therefore eliminating the text's content. Text classification methods were tested on separate cell devices to see how well they filtered SMS spam by Nuruzzaman et al. [6]. All learning, screening, and updating operations were all carried out on a separate cell phone. Their suggested model was able to sort SMS spam with some excellent accuracy and low memory 20 N. Nuruzzaman et al. [6] have made significant utilization, and sufficient batch processing without requiring a huge dataset or any machine help.

Jayasingh et al. [7] have done weather prediction using hybrid soft computing models. Uysal et al. [8] presented a technique for SMS spam filtering that employs different feature selection methodologies, namely chi-square metrics and mutual information, to select information. They have created a real-time smartphone app for SMS spam detection based on the android platform. They utilized two Bayesian-based classification methods, Bayesian and binary. According to the authors, their suggested method detects both spam and genuine communications with great accuracy. For SMS spam detection, Yadav et al. [9] created a model text Assassin. They utilized a minimal feature set of 20 features as well as two machine learning algorithms, support vector machine (SVM) and Bayesian learning. They gathered a total of 2000 messages from visitors over the course of two months. In their suggested approach, anytime a user receives a message on mobile phone, SMS Assassin collects it without the user's awareness, retrieves feature values, and transmits them to the server. If the communications are marked as spam, the user will not be able to access them and they will be sent to the spam folder. The use of Bayesian filtering to identify SMS spam has been investigated by Hidalgo et al. [10]. They created two sets of data, one in English and the other in Spanish. According to their findings, Bayesian filters technologies that were previously used to identify spam emails may now be used to stop SMS spam. Cormack et al. [11] discussed feature engineering for mobile spam filtering. Jun et al. [12] explained the spam detection approach for secure mobile message communication using machine learning algorithms. Cormack et al. [13] explained the different types of machine learning algorithm. Jayasingh et al. [14] showed a novel approach for data classification using neural network.

3 Proposed Method

The suggested investigation in this work involves the procedure depicted in Fig. 2. The basic goal is to categorize text messaging either as ham or as spam.

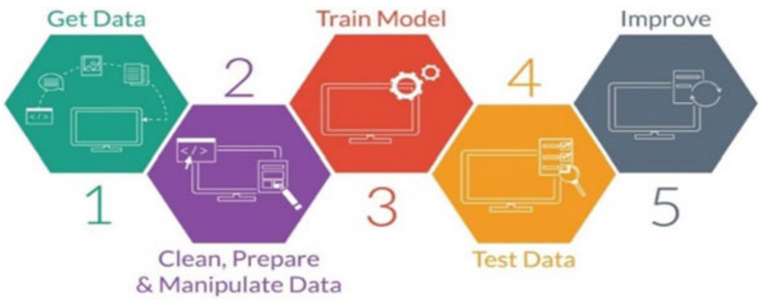
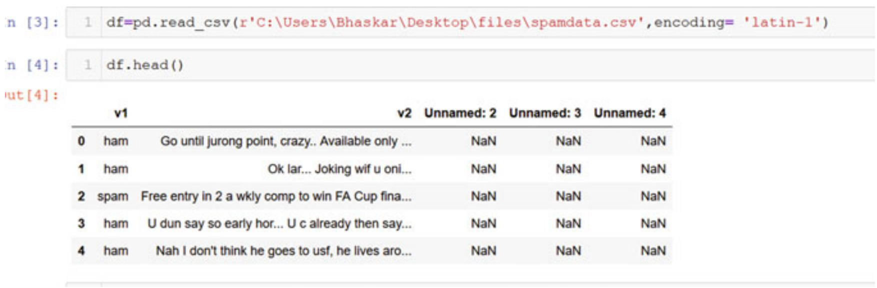


Fig. 2 Overview of approach for the model



```
n [3]:  1 df=pd.read_csv(r'C:\Users\Bhaskar\Desktop\files\spamdata.csv',encoding= 'latin-1')
n [4]:  1 df.head()
ut[4]:
```

	v1	v2	Unnamed: 2	Unnamed: 3	Unnamed: 4
0	ham	Go until jurong point, crazy.. Available only ...	NaN	NaN	NaN
1	ham	Ok lar... Joking wif u oni...	NaN	NaN	NaN
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...	NaN	NaN	NaN
3	ham	U dun say so early hor... U c already then say...	NaN	NaN	NaN
4	ham	Nah I don't think he goes to usf, he lives aro...	NaN	NaN	NaN

Fig. 3 Sample dataset before cleaning

4 Dataset

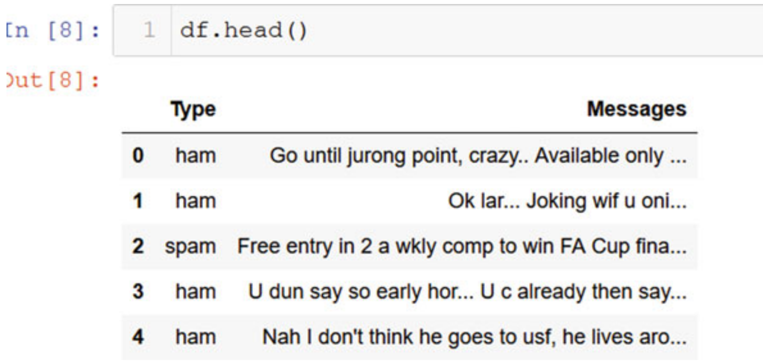
We have taken the dataset having 5574 tagged data in the sample which contains 4827 ham messages and 747 spam messages. The dataset has two columns named ham and spam. It also has three unnamed columns as shown in Fig. 3.

Preprocessing, feature extraction, selection, and classification are all steps of spam detection as shown in Fig. 4.

5 Methodology

Our approach's major goal is to categories spam text messages as soon as they arrive on a smartphone, independently of whether they are newly produced spam messaging. We began by gathering data and finalizing the features for our trial. After finalizing the characteristics, we created a feature representation by extracting features from the messages (ham and spam). Several extracted features are used for training and testing. Our suggested system bases its judgment on ten characteristics.

The control structure of our suggested method is depicted. During the training process, the extracted features of phishing and ham texts are used to create a classifier



In [8]:
```
df.head()
```

Out[8]:

	Type	Messages
0	ham	Go until jurong point, crazy.. Available only ...
1	ham	Ok lar... Joking wif u oni...
2	spam	Free entry in 2 a wkly comp to win FA Cup fina...
3	ham	U dun say so early hor... U c already then say...
4	ham	Nah I don't think he goes to usf, he lives aro...

Fig. 4 Sample dataset after cleaning

model. The classification evaluates if a new text is spam or not during the testing stage. Finally, classification of data for several machine learning techniques is done, and performance for each machine learning algorithm is assessed in order to choose the optimum methodology for our suggested strategy processing is the initial step inside the conversion of unstructured information to much more structured information, because characters are frequently substituted for terms in text messages. The stop word list extractor for English language, used in this research to eliminate stop words from SMS text messages depicts the frequency of terms in SMS messages, whereas the most familiar terms used in spam text messages belong to the noun and predicate categories. Similarly, pronouns, proposition, and a variety of many other stop words dominate the top terms in ham SMS messages.

6 Data Preprocessing

First, we converted the raw data into meaningful data. Then, we removed null values from our dataset, and then, we labeled the data into spam and ham. We lowered all the words and used NLTK stop words to remove the repeated word to reduce the noise. Then, we used porters Temer method to convert the word into vectors which have many dimensions because there are a lot of different words present in our dataset.

7 Algorithms

KNN algorithm

It is a supervised algorithm which can be used for regression and classification problems. Therefore, in businesses, it is mostly utilized to solve classification and prediction issues. The two following characteristics will be a good way to describe KNN:

- KNN is a lazy learning algorithm since it does not have a learning and development period and instead utilizes all of the labeled training data and classification.
- KNN would also be a nonparametric learning method since it makes no assumptions regarding data.

Random forest Algorithms

It is a supervised algorithm which is used for classification problems. It mainly comprises of trees to give rise to a forest. The random forest constructs decision trees using datasets, finds the prediction in each, and votes for the best prediction. It is a based on integrated that is superior than just a decision tree classifier since it averages the results to minimize overfitting.

Naive Bayes Algorithms

The Bayes theorem is used in naive Bayes algorithms, which is a classification approach which focuses on the strong hypothesis that all predictions are autonomous of one another. A smartphone, for example, may be deemed intelligent if it has a touchscreen interface, Internet connectivity, and an excellent camera. Despite the fact that all of these characteristics are interdependent, they individually add to the likelihood that the smartphone.

Decision tree algorithm

An algorithmic technique that can partition the information in various ways based on different circumstances can be used to create decision trees. The much more powerful algorithm in the field of supervised learning algorithms is decision tree.

This can also be used for classification as well as regression. The decision nodes, in which the input is split, as well as the leaves, where we can get the result, are the two major components of a tree. A binary tree is used to forecast whether a person is healthy or unfit based on numerous factors such as age, diet and lifestyle, and daily exercise.

Confusion Matrix

The confusion matrix provides the details of prediction in a model as shown in Fig. 5. The **true positive** indicates that a positive value is predicted correctly [1]. A **false positive** occurs when the model forecasts the positive class erroneously as depicted. False positive indicates that a negative value is predicted as positive [0, 1]. **False negative** indicates that a positive value is predicted as negative [1, 0]. Always FN

Fig. 5 Confusion matrix

TN	FN
FP	TP

should be less in our model. **True negative** indicates that values that are actually negative and predicted also negative [0, 0].

Accuracy: It is the proportion of times the model correctly predicted the value. (AKA, the percentage of situations in which an industry's success or failure was correctly predicted). At first sight, this appears to be an excellent success statistic because the model was right the vast majority of the time. However, it is dependent on the situation.

The accuracy is defined by division of true positive + true negative by the sum of true positive + true negative + false positive + false negative.

Recall: It is the division of the true positive values by true positive and false negative values.

$$\text{Recall} = \text{TP}/\text{TP} + \text{FN}$$

Precision: It indicates the percentage of spam messages that the classification system correctly identified as spam. It demonstrates that everything is correct.

$$\text{Precision} = \text{TP}/\text{TP} + \text{FP}$$

8 Analysis

The execution of the proposed SMS spam detection system is evaluated through a series of tests. We first chose feature extraction methods on spam and ham message behavior and then separated this information from the data to create the classification model. To get system performance, different classification algorithms are used. They are support vector machine (SVM), random forest, AdaBoost, and decision tree.

It provides a built-in solution to cope with this issue. During split discovery, it overlooks missing data and then assigns them to the side that reduces the loss the greatest. If we do not want to utilize option and instead use alternative approaches, change the parameter to: Before starting the model prediction, we need to analyze the words, which words are repeated. The most common words are there in SMS, which are shown in Fig. 6. So, again we need to find common words in spam messages as shown in Fig. 7 and ham messages as shown in Fig. 8.

We have to find out the data which contains the information with a small number of potential values. Machine learning algorithms typically operate directly with categorical attributes, and in order to use them, subcategories must first be converted to integers before the deep learning model can be applied to them.

One-hot-encoder and updating values are two methods that may be used. To convert category variables into dummy/indicator variables, pandas is used to get dummy function. Imputation and erasure are two strategies for dealing with missing

```
In [233]:
wordcloud=WordCloud(background_color='white',width=1000,height=1000,max_words=50).generate(str(df['Messages']))
plt.rcParams['figure.figsize']=(10,10)
plt.title('Most common number of words in Messages')
plt.axis('off')
plt.imshow(wordcloud)
```

Fig. 6 Most common number of words in the message

```
In [234]:
wordcloud=WordCloud(background_color='white',width=1000,height=1000,max_words=50).generate(str(Spamfilter))
plt.rcParams['figure.figsize']=(10,10)
plt.title('Most common number of words in Messages')
plt.axis('off')
plt.imshow(wordcloud)
```

Fig. 7 Most common number of words in the spam message

```
In [235]:
wordcloud=WordCloud(background_color='white',width=1000,height=1000,max_words=50).generate(str(Hamfilter))
plt.rcParams['figure.figsize']=(10,10)
plt.title('Most common number of words in Messages')
plt.axis('off')
plt.imshow(wordcloud)
```

Fig. 8 Most common number of words in the ham message

values. Feature scaling is some of the most essential phases in the preprocessing of data before constructing a machine learning technique in machine learning algorithms. Many times, the dataset will contain features with a wide range of orders of magnitude, units, and ranges. Using unoptimized features, on the other hand, creates a difficulty in most machine learning applications. Machine learning methods that employ gradient descent as an optimization procedure, such as classification model and neural networks, require the dataset to be scaled. Whenever we employ proximity algorithms like random forest and SVM, which are most impacted by the set of advantages, we run into a similar dilemma. The reason for this is that they use data point distances to evaluate how similar they are. We do not have to be concerned with its feature scalability. The size of the features has little effect on tree-based algorithms. The comparison of different algorithms is shown in Table 1.

It is observed from Table 1 and Fig. 9 that we have obtained the most accurate result by using random forests. Furthermore, using the random forest machine learning method, we were able to obtain 99.9% accuracy as shown in the graphical comparison of accuracy of different models in Fig. 9. During the training phase, the random forest classification method creates a collection of decision trees, which are subsequently used to classify randomly picked characteristics. The classification of ham and spam messages is done using the random forest (RF) algorithm. Random forest algorithm makes use of decision trees and eliminates the overfitting problem of the decision trees. Each tree present in the random forest algorithm provides its prediction, which vary from one another. The average of their performances is calculated. In the training

Table 1 Comparison of different algorithms on the basis of accuracy, precision, recall, and Roc-curve

Algorithms	Accuracy	Precision	Recall	Roc-curve
Random forest	99.99	99.98	99.99	99.89
Naïve Bayes	98.58	98.44	98.58	98.54
Decision tree	94.93	94.69	94.93	94.12
AdaBoost	92.98	92.53	92.98	92.54
SVM	90.59	90.27	90.59	90.11
KNN	96.01	96.00	96.01	95.89

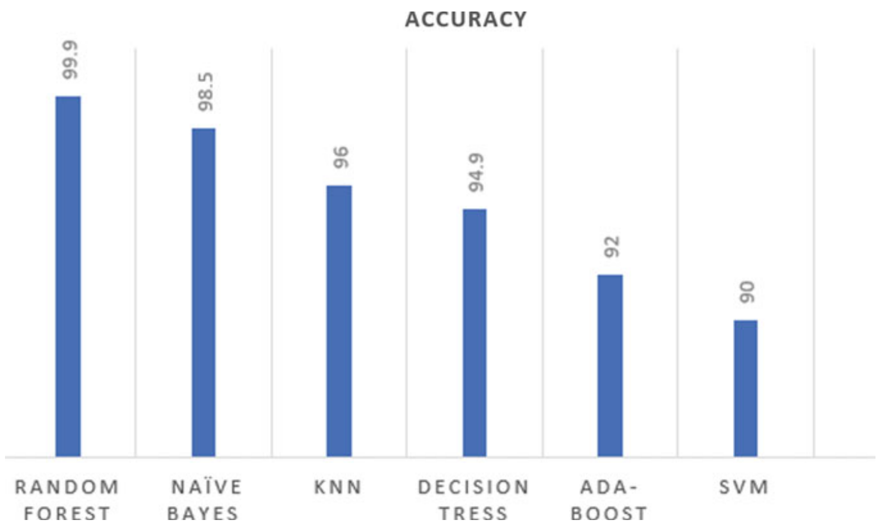


Fig. 9 Comparison of different algorithms on the basis of accuracy

phase, many decision trees are created and then they operate on a set of randomly selected features.

9 Conclusion and Future Work

As there is increase in usage of text messaging day by day, the problem of SMS spam is becoming more prevalent. Filtering SMS spam has been a major problem in recent years. In this research work, we have used six different machine learning algorithms such as decision tree, naive Bayes, random forest, KNN, SVM, and AdaBoost in which ten features from the dataset are used. There are 5574 tagged data in the sample, including 4827 messages belonging to ham messages and 747 messages belonging to spam messages collected. With accuracy of 99.9%, the random forest classification algorithm outperforms all other classification algorithms. The accuracy may further be enhanced by using hybrid soft computing models.

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